

# **Mathematics Bootcamp**

## **Optimization and Consumer Demand**

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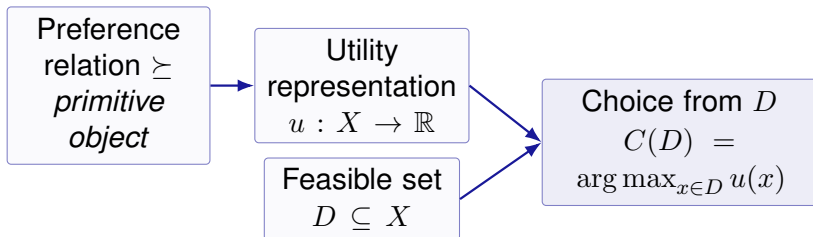
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# Today

- Existence
  - Maximum versus argmax; failure and the Extreme Value Theorem
- Candidates and Verification
  - First-order conditions; curvature and concavity
- Constraints
  - Lagrange multipliers; complementary slackness and Kuhn–Tucker
- Consumer Demand
  - Cobb–Douglas in full; corners and kinks as contrasts

# Optimization Solves the Argmax from Day 1

Day 1 connected preferences and observed choice through rationalization. We now study the mathematical problem inside that connection.



## The Mathematical Question

Which points in  $D$  maximize the objective, and what assumptions let us find and justify them?

# The Maximum and the Maximizers Are Different Objects

Consider

$$\max\{f(x) \mid x \in D\}.$$

## Maximum Value

If a solution exists, then

$$\max\{f(x) \mid x \in D\} = f(x^*) \in \mathbb{R}.$$

This is a number.

## Maximizers

$$\begin{aligned} & \arg \max\{f(x) \mid x \in D\} \\ &= \{x \in D : f(x) \geq f(y) \ \forall y \in D\}. \end{aligned}$$

This is a set of choices.

The argmax may be empty, contain one point, or contain many points.

# A Maximizer Can Fail to Exist in Three Different Ways

## Open feasible set

$$\max_{x \in (0,1)} x$$

Supremum 1, but no maximizer.

## Unbounded objective

$$\max_{x \geq 0} x$$

No finite maximum.

## Discontinuity

$$f(x) = \begin{cases} x, & x < \frac{1}{2}, \\ x - 1, & x \geq \frac{1}{2}. \end{cases}$$

On  $[0, 1]$ , the supremum is not attained.

## Three Distinct Failures

The feasible set may omit its boundary, the objective may be unbounded, or the objective may fail to preserve limits. In each case the argmax is empty.

# The Extreme Value Theorem Supplies an Existence Checklist

In  $\mathbb{R}^n$ , a set is compact if and only if it is closed and bounded.

## Extreme Value Theorem

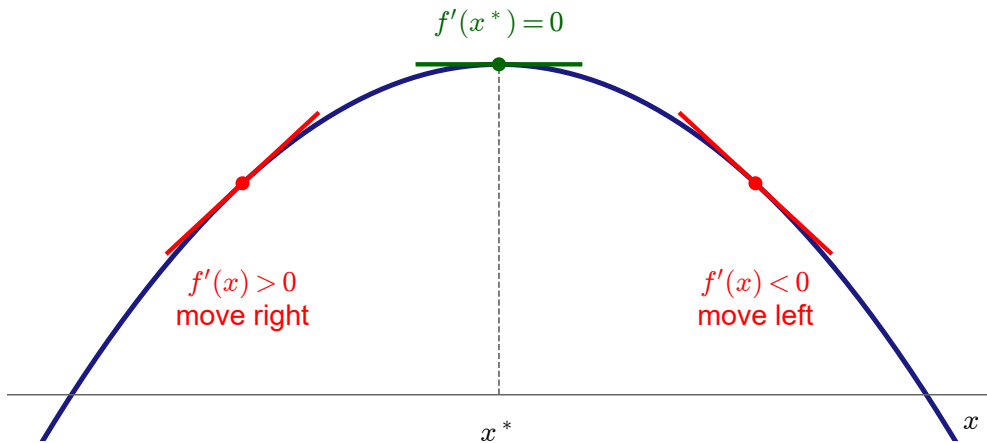
If  $D \subseteq \mathbb{R}^n$  is nonempty and compact and  $f : D \rightarrow \mathbb{R}$  is continuous, then  $f$  attains a maximum and a minimum on  $D$ .

Before calculating, verify four separate facts:

1.  $D$  is nonempty;
2.  $D$  is closed;
3.  $D$  is bounded;
4.  $f$  is continuous on  $D$ .

**These assumptions guarantee existence, not uniqueness.**

## If the Slope Is Nonzero, Improvement Is Possible



# Why the First-Order Condition Is Necessary

## First-Order Necessary Condition

If  $x^* \in \text{int } D \subseteq \mathbb{R}$  is a local maximizer of  $f$  on  $D$  and  $f$  is differentiable at  $x^*$ , then  $f'(x^*) = 0$ .

(1) For sufficiently small  $h > 0$ , local optimality gives

$$f(x^* + h) \leq f(x^*) \implies \frac{f(x^* + h) - f(x^*)}{h} \leq 0.$$

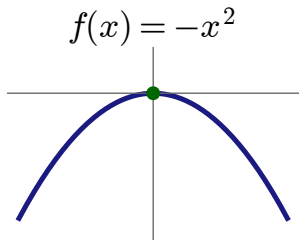
(2) For sufficiently small  $h < 0$ , the numerator is still nonpositive but the denominator is negative, so

$$\frac{f(x^* + h) - f(x^*)}{h} \geq 0.$$

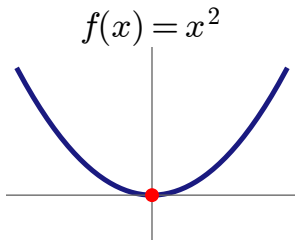
(3) The two one-sided limits are equal because  $f$  is differentiable. Hence  $f'(x^*) \leq 0$  and  $f'(x^*) \geq 0$ , so  $f'(x^*) = 0$ .



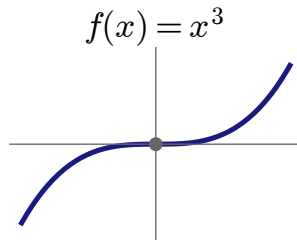
## A Critical Point Is Only a Candidate



local and global  
maximum



local and global  
minimum



neither

All three satisfy  $f'(0) = 0$ . Moreover, a boundary maximizer need not satisfy  $f'(x) = 0$  because movement may be possible in only one direction.

# The Gradient Tests Every Direction at Once

For a direction  $d \in \mathbb{R}^n$ ,

$$f(x + td) - f(x) \approx t \nabla f(x) \cdot d.$$

If  $x^*$  is an interior local maximum, define

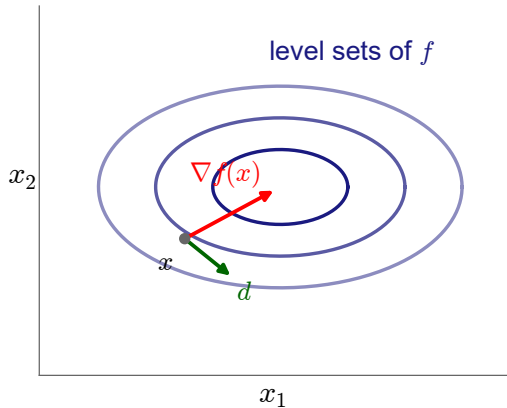
$$\phi_d(t) = f(x^* + td).$$

Then  $t = 0$  locally maximizes  $\phi_d$ , so

$$\phi'_d(0) = \nabla f(x^*) \cdot d = 0 \quad \text{for every } d.$$

Therefore,

$$\boxed{\nabla f(x^*) = 0.}$$



# Curvature Turns Candidates into Conclusions

## Local Conclusion

Suppose  $f$  is twice differentiable and

$$\nabla f(x^*) = 0.$$

If  $\nabla^2 f(x^*)$  is negative definite, then  $x^*$  is a strict local maximizer.

Negative semidefiniteness at  $x^*$  is necessary for a smooth local maximum, but may not be sufficient.

First-order conditions generate candidates; curvature and concavity tell us what those candidates mean.

## Global Conclusion

Suppose  $D$  is convex and  $f$  is differentiable and concave on  $D$ . Then

$$\nabla f(x^*) = 0 \implies x^* \in \arg \max \{f(x) \mid x \in D\}.$$

If  $f$  is strictly concave, the maximizer is unique whenever it exists.

# A Constraint Removes Directions

Consider

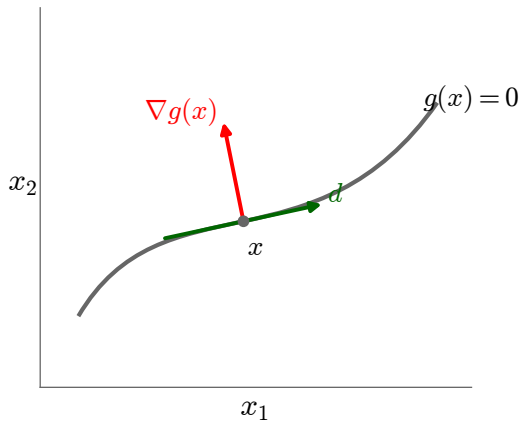
$$\max_x f(x) \quad \text{subject to} \quad g(x) = 0.$$

A direction  $d$  tangent to the constraint satisfies, to first order,

$$\nabla g(x) \cdot d = 0.$$

At a constrained optimum, the objective must not increase along any such feasible direction.

The gradient  $\nabla g(x)$  is normal to the constraint curve.



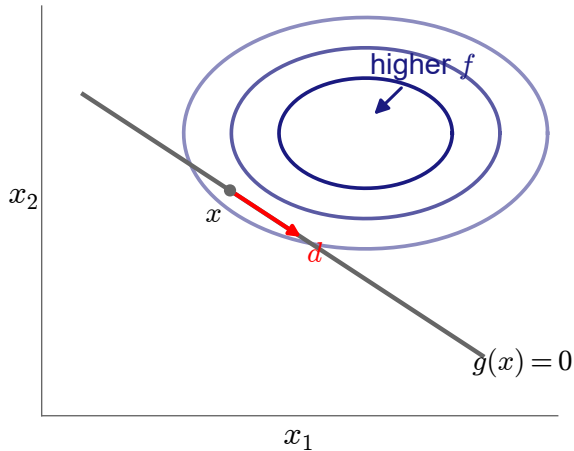
## If the Curves Cross, Improvement Remains Possible

If the feasible curve crosses an objective level curve, one of the two feasible directions leads toward a higher level.

At the marked point,

$$\nabla f(x) \cdot d > 0.$$

A small feasible movement in direction  $d$  raises the objective, so the point cannot be optimal.



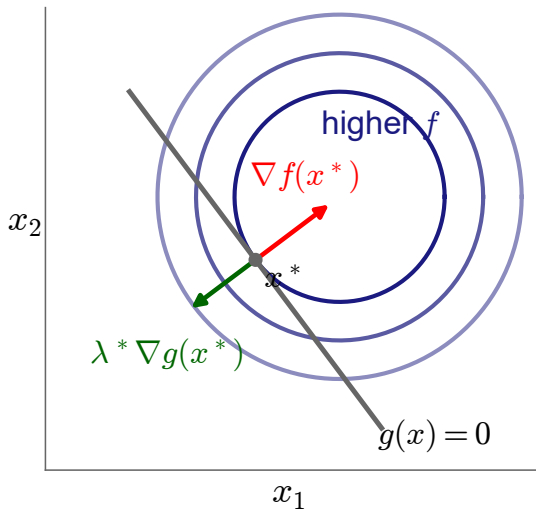
# At a Smooth Constrained Optimum, the Normals Align

At tangency, the objective level curve and the feasible curve share the same tangent direction. Their normal vectors must therefore be parallel:

$$\nabla f(x^*) + \lambda^* \nabla g(x^*) = 0.$$

Equivalently, every feasible tangent direction  $d$  satisfies

$$\nabla f(x^*) \cdot d = 0.$$



# The Constraint Cuts a Feasible Curve from the Surface

Consider

$$f(x) = 4 - [(x_1 - 1)^2 + (x_2 - 1)^2],$$

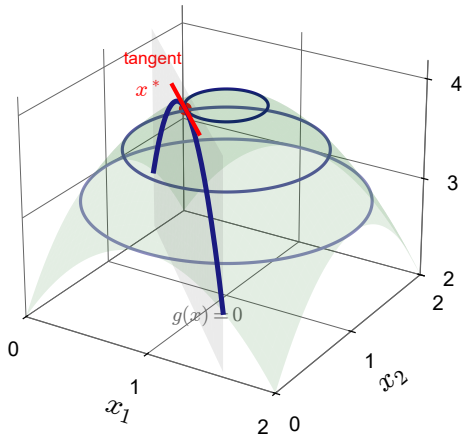
$$g(x) = x_1 + x_2 - 1.6 = 0.$$

The gray plane cuts a blue feasible curve from the green objective surface. Maximizing  $f$  on  $g(x) = 0$  means choosing its highest point:

$$x^* = (0.8, 0.8).$$

At  $x^*$  the red tangent is horizontal:

$$d = (1, -1), \quad \nabla f(x^*) \cdot d = 0.$$



# The Lagrangean Packages Tangency and Feasibility

For

$$\max_x f(x) \quad \text{subject to} \quad g(x) = 0,$$

define

$$L(x; \lambda) = f(x) + \lambda g(x).$$

## Stationarity

$$\begin{aligned}\nabla_x L(x^*; \lambda^*) &= 0, \\ \nabla f(x^*) + \lambda^* \nabla g(x^*) &= 0.\end{aligned}$$

## Feasibility

$$g(x^*) = 0.$$

The candidate must satisfy the constraint.

These equations generate candidates. Concavity or direct comparison is still needed to establish global optimality.



# Regular Constraints Make the Multiplier Well Defined

## Theorem of Lagrange

Let  $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$  be continuously differentiable and let  $U$  be open. If  $x^*$  is a local optimum of  $f$  on  $D = U \cap \{x : g(x) = 0\}$  and  $\nabla g(x^*) \neq 0$ , then there exists a unique  $\lambda^* \in \mathbb{R}$  such that

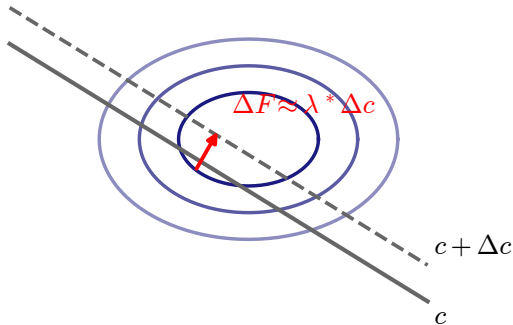
$$\nabla f(x^*) + \lambda^* \nabla g(x^*) = 0.$$

- The condition  $\nabla g(x^*) \neq 0$  rules out a locally degenerate boundary.
- It guarantees existence and uniqueness of the multiplier at that fixed  $x^*$ .
- It does not guarantee that the optimizer itself is unique.

# A Multiplier Measures the Value of Relaxing a Constraint

Let  $g(x; c) = g(x) + c$ , let  $x^*(c)$  solve  $\max\{f(x) \mid g(x; c) = 0\}$ , and define  $F(c) = f(x^*(c))$ . If  $x^*(c)$  varies smoothly, then  $F'(c) = \lambda^*(c)$ .

- $\lambda^*$  is first a coefficient aligning two normal vectors.
- With the additional smoothness, it is the marginal optimized value of relaxing the constraint.
- Its value depends on the chosen representation of the objective.



# Inequality Constraints Are Effective or Slack

Write each inequality as  $h_i(x) \geq 0$ . At a candidate  $x^*$ :

## Slack Constraint

$$h_i(x^*) > 0.$$

The constraint is slack. Small changes in its boundary do not affect the local choice, so  $\lambda_i^* = 0$ .

## Effective Constraint

$$h_i(x^*) = 0.$$

The boundary restricts feasible directions and may carry a positive multiplier.

Both cases are summarized by

$$\lambda_i^* \geq 0, \quad h_i(x^*) \geq 0, \quad \lambda_i^* h_i(x^*) = 0.$$

# Kuhn–Tucker Adds One-Sided Feasibility

Consider

$$\max_x f(x) \quad \text{subject to} \quad h_i(x) \geq 0, \quad i = 1, \dots, m,$$

and define  $L(x; \lambda) = f(x) + \sum_{i=1}^m \lambda_i h_i(x)$ .

## Kuhn–Tucker Conditions

Under a constraint qualification, a smooth local maximizer has multipliers  $\lambda_i^* \geq 0$  satisfying

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*) = 0,$$

$$h_i(x^*) \geq 0, \quad \lambda_i^* \geq 0, \quad \lambda_i^* h_i(x^*) = 0.$$

Slack constraints have zero multipliers; effective constraints may have positive multipliers. If  $f$  and the  $h_i$  are concave, these conditions also verify global optimality.

## Time for Economics

- The objective  $f$  becomes a utility representation  $u$ .
- The feasible set  $D$  becomes the budget set  $B(p, I)$ .
- The general first-order conditions become restrictions on marginal utility per unit of money.
- Corners and kinks determine when ordinary tangency is the wrong method.

### The Same Mathematical Question

Which affordable bundle lies on the highest attainable indifference curve?

# Consumer Choice Is the Same Optimization Problem

A consumer chooses  $x \in \mathbb{R}_+^L$ , has income  $I > 0$ , and faces prices  $p \in \mathbb{R}_{++}^L$ .

## Walrasian Budget Set

$$B(p, I) = \{x \in \mathbb{R}_+^L : p \cdot x \leq I\}.$$

## Walrasian Demand Correspondence

$$x(p, I) = \arg \max \{u(x) \mid x \in B(p, I)\}.$$

A selected optimum is written  $x^* \in x(p, I)$ .

Demand is a correspondence because several bundles may maximize utility.

## The Assumptions from Day 3 Now Do Specific Jobs

| Assumption                   | Consequence for consumer choice                            |
|------------------------------|------------------------------------------------------------|
| $p \gg 0$ and $I > 0$        | $B(p, I)$ is nonempty, compact, and convex.                |
| Continuity of $u$            | $x(p, I)$ is nonempty and compact.                         |
| Strict quasiconcavity of $u$ | At most one bundle maximizes $u$ on the convex budget set. |
| Local nonsatiation           | Every optimal bundle exhausts the budget.                  |

These are distinct conclusions: existence of a maximizer, uniqueness of the maximizer, and existence of multipliers are different questions.

# Local Nonsatiation Makes the Budget Bind

## Walras' Law for Individual Demand

If preferences are locally nonsatiated and  $x^* \in x(p, I)$ , then

$$p \cdot x^* = I.$$

- (1) Suppose instead that  $p \cdot x^* < I$ . There is positive spending slack.
- (2) Because prices are finite, every sufficiently close bundle remains affordable.
- (3) Local nonsatiation gives a sufficiently close bundle  $y$  with  $y \succ x^*$ .
- (4) Then  $y \in B(p, I)$  and  $y \succ x^*$ , contradicting  $x^* \in x(p, I)$ .

Hence the inequality constraint may be treated as the equality  $p \cdot x = I$  when solving a locally nonsatiated consumer problem.



## A Money-Neutral Reallocation Is a Feasible Direction

Move  $\delta > 0$  units of money from good  $m$  to good  $\ell$ :

$$\Delta x_\ell = \frac{\delta}{p_\ell}, \quad \Delta x_m = -\frac{\delta}{p_m}.$$

The change in expenditure is zero:

$$p_\ell \Delta x_\ell + p_m \Delta x_m = \delta - \delta = 0.$$

To first order, the utility change is

$$\Delta u \approx \delta \left( \frac{\partial u(x)/\partial x_\ell}{p_\ell} - \frac{\partial u(x)/\partial x_m}{p_m} \right).$$

If the expression in parentheses is positive, the current bundle is not optimal.

## At an Interior Optimum, No Money Switch Pays

If goods  $\ell$  and  $m$  are consumed in strictly positive amounts, both reallocation directions are feasible. Therefore,

$$\boxed{\frac{\partial u(x^*)/\partial x_\ell}{p_\ell} = \frac{\partial u(x^*)/\partial x_m}{p_m} = \lambda^*}.$$

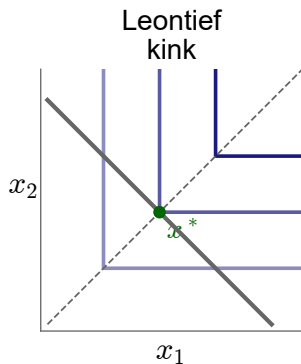
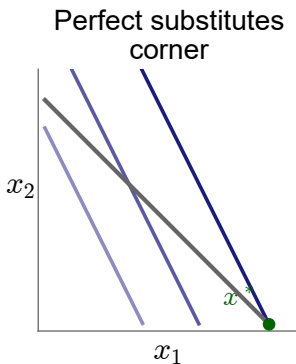
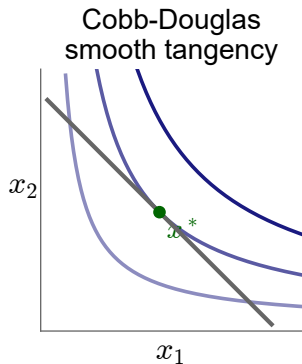
Equivalently,

$$\nabla u(x^*) = \lambda^* p, \quad \frac{\partial u(x^*)/\partial x_\ell}{\partial u(x^*)/\partial x_m} = \frac{p_\ell}{p_m}.$$

### Interpretation

Each consumed good produces the same marginal utility per unit of money. Under the smooth value-function assumptions,  $\lambda^* = \partial v(p, I)/\partial I$ .

# The Shape of Indifference Curves Determines the Method



**The Budget Is the Same; the Utility Geometry Changes**

**Smooth:** tangency

**Linear:** corner

**Fixed proportions:** kink

## A Monotone Transformation Simplifies Cobb–Douglas

Let

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, \quad 0 < \alpha < 1.$$

Because  $\log$  is strictly increasing,

$$\arg \max \{u(x) \mid x \in B(p, I)\} = \arg \max \{\alpha \log x_1 + (1 - \alpha) \log x_2 \mid x \in B(p, I)\}.$$

With local nonsatiation, use the binding budget and write

$$L(x; \lambda) = \alpha \log x_1 + (1 - \alpha) \log x_2 + \lambda(I - p_1 x_1 - p_2 x_2).$$

The maximizing bundle is unchanged by the transformation; the numerical value of the multiplier is not.

## Cobb–Douglas Demand Allocates Fixed Shares of Income

The first-order conditions are

$$\frac{\alpha}{x_1} = \lambda^* p_1, \quad \frac{1 - \alpha}{x_2} = \lambda^* p_2, \quad p_1 x_1 + p_2 x_2 = I.$$

Multiplying the first two equations by  $x_1$  and  $x_2$  gives

$$\alpha = \lambda^* p_1 x_1, \quad 1 - \alpha = \lambda^* p_2 x_2.$$

Summing and using the budget yields  $\lambda^* = 1/I$ , so

$$x_1(p, I) = \frac{\alpha I}{p_1}, \quad x_2(p, I) = \frac{(1 - \alpha) I}{p_2}.$$

The consumer spends the shares  $\alpha$  and  $1 - \alpha$  of income on the two goods.

## Perfect Substitutes Turn Tangency into a Comparison

Let  $u(x_1, x_2) = ax_1 + bx_2$ , where  $a, b > 0$ . Marginal utility per unit of money is constant:

$$\frac{a}{p_1} \quad \text{and} \quad \frac{b}{p_2}.$$

### Walrasian Demand

$$x(p, I) = \begin{cases} \{(I/p_1, 0)\}, & a/p_1 > b/p_2, \\ \{(0, I/p_2)\}, & a/p_1 < b/p_2, \\ \{x \in \mathbb{R}_+^2 : p \cdot x = I\}, & a/p_1 = b/p_2. \end{cases}$$

Unless the two returns are equal, the optimum is a corner rather than a tangency.

# Leontief Utility Is Solved at the Kink

Let

$$u(x_1, x_2) = \min\{ax_1, bx_2\}, \quad a, b > 0.$$

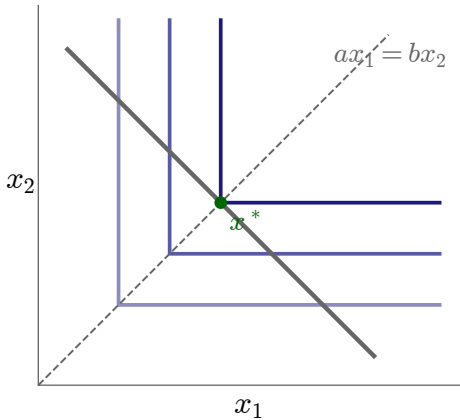
Unmatched units do not raise utility, so an optimum satisfies

$$ax_1 = bx_2.$$

Together with  $p_1x_1 + p_2x_2 = I$ ,

$$x_1(p, I) = \frac{bI}{bp_1 + ap_2}, \quad x_2(p, I) = \frac{aI}{bp_1 + ap_2}.$$

Ordinary smooth tangency conditions do not apply at the economically relevant kink.



# Kuhn–Tucker for the Consumer Problem

Write the constraints as  $h_\ell(x) = x_\ell$  for  $\ell = 1, \dots, L$  and  $h_{L+1}(x) = I - p \cdot x$ . With nonnegative multipliers,

$$L(x; \lambda) = u(x) + \sum_{\ell=1}^L \lambda_\ell x_\ell + \lambda_{L+1}(I - p \cdot x).$$

With local nonsatiation, the budget binds. For every good, stationarity and complementary slackness give

$$\frac{\partial u(x^*)}{\partial x_\ell} + \lambda_\ell^* - \lambda_{L+1}^* p_\ell = 0, \quad \lambda_\ell^* x_\ell^* = 0.$$

If  $x_\ell^* > 0$ , then  $\lambda_\ell^* = 0$  and

$$\frac{\partial u(x^*)/\partial x_\ell}{p_\ell} = \lambda_{L+1}^*.$$

If  $x_\ell^* = 0$ , then

$$\frac{\partial u(x^*)/\partial x_\ell}{p_\ell} \leq \lambda_{L+1}^*.$$



# Every Case Asks the Same Question

## Central Question

Is there a feasible direction that raises the objective?

1. Write the objective and the feasible set.
2. Establish existence and uniqueness separately.
3. Identify whether the candidate is interior, on a smooth boundary, at a corner, or at a kink.
4. Generate candidates using first-order conditions, the Lagrangean method, Kuhn–Tucker, or geometry.
5. Verify feasibility, complementary slackness, and global optimality.

**First-order conditions reduce the search; assumptions and verification finish it.**